Visually Impaired AI Navigation Assistant System

Team Name

Cheongju Eagles

Region

Region 6: Eastern Asia + South Eastern Asia

Category

Visually impaired assistance.

Team Type

University Team

Laboratory details: Multi-media image processing lab, Department of information and Communication, School of Electrical and Computer engineering, Chungbuk National University, South Korea.

Team Members

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- **Biography of team members**

Hatem Ibrahem received his B.Eng. degree in electrical engineering (electronics and communication) from Assiut University, Assiut, Egypt, in 2013. He is currently pursuing the combined master's and Ph.D degree with the School of Information and Communication Engineering, Chungbuk National University, Chungbuk, South Korea.
His research interests include multimedia, image processing, machine learning, deep learning and computer vision.
Ahmed Salem received the B.Eng. degree in electrical engineering (electronics & communication) from Assiut University, Assiut, Egypt, in 2012, and the M.Eng. degree in electronics and communication engineering from Egypt-Japan University of Science and Technology, Alexandria, Egypt, in 2016.
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Bilel Yagoub received his B.Sc degree in computer science from University Abdel-hamid Ibn Badis Mostaganem, Algeria, in 2013, and the M.Sc degree from the University Oran 1 Ahmed Ben Bella, Algeria, in 2015. He is currently pursuing the Ph.D degree with the school of Information and Communication Engineering, Chungbuk National University, Chungbuk, South Korea. His research interests include Web/Mobile application development, deep learning, and computer vision.
Hyun-Soo Kang received the BS degree in electronic engineering from Kyoungpook National University, Republic of Korea, in 1991, and the MS and PhD degrees in electrical and electronics

engineering from KAIST in 1994 and 1999, respectively. From 1999 to 2005, he was with Hynix Semiconductor Co., Ltd., Electronics and Telecommunications Research Institute (ETRI), and Chungang University. He joined the College of Electrical and Computer Engineering of Chungbuk National University, Chungbuk, Republic of Korea, in Mar. 2005.
His research interests include image compression and image processing.

Team Capability

- **Team experiences**

Since our lab members have experiences with image processing and deep learning, we can provide all the technical work mentioned in this report as we experiences and practiced this work in our researches and studies. Hatem Ibrahem worked in object detection of street scene in his study and also worked in semantic segmentation for general objects. Ahmed Salem worked in light field 3D reconstruction and depth estimation as it is his main focus. Bilel Yagoub has a background from computer science, so he is experienced in different platform application development, he did several projects of Mobile applications, desktop application and hybrid-platform applications, and he also integrated computer vision models in some of those applications. Prof. Hyun Soo Kang has a long experience in image processing and computer vision and he will provide the guidance for the project members.

- **Publications (recent 3 years)**

(1) H. Ibrahem, A. Salem, H. Kang, "Real-Time Weakly-Supervised Object Detection Using Center-of-Features Localization," (Under-review) IEEE Access journal, 2021
(3) A. Salem, H. Ibrahem, B. Yagoub, H. Kang,. "Stereo Extrapolation: View synthesis with multiplane images". (Accepted paper but not published yet). IEEE ICEIC conference, 2021
(2) H. Ibrahem, A. Salem, B. Yagoub, H. Kang. "CCC: Weakly Supervised Semantic Segmentation Using Class Contribution Classification". (Accepted paper but not published yet). IEEE ICEIC conference, 2021
(4) B. Yagoub, H. Ibrahem, A. Salem, J. Suh, H. Kang,. "X-ray image Denoising for Cargo Dual Energy Inspection System". (Accepted paper but not published yet). IEEE ICEIC conference, 2021
(5) A. Salem, H. Ibrahem, H. Kang, "Fast Light Field Reconstruction Using Convolutional Neural Network to Double Angular Resolution," IEEE ICCE, 2020
(6) H. Ibrahem, A. Salem, H. Kang, "Weakly Supervised Traffic Sign Detection in Real Time Using Single CNN Architecture for Multiple Purposes," IEEE ICCE, 2020
(7) A. Salem, H. Ibrahem, H. Kang, " Dual Disparity-Based Novel View Reconstruction for Light Field Images Using Discrete CosineTransform Filter," IEEE Access Journal, 2020
(8) H. Choi, H. Kang, B. Yun, "CNN-Based Illumination Estimation with Semantic Information," Applied Sciences, 2020.
(9) H. Choi, H. Kang, B. Yun, " Tone Mapping of High Dynamic Range Images Combining Co-Occurrence Histogram and Visual Salience Detection," Applied Sciences, 2019.
(10) H. Choi, H. Kang, " Tone Mapping for High Dynamic Range Image Using Modified Guided Filter and TV-based Restoration Model," Journal of Imaging Science and Technology, 2018

- **Projects (recent 3 years)**

(1) "Super-resolution deep learning network development for light field images" funded by Korea Research Foundation, 2020~2023 (3 years)

(2) "View synthesis and depth estimation for light field images" funded by Korea Research Foundation, 2017~2020 (3 years)
(3) "Traffic sign detection and recognition" funded by Korea Railroad Research Institute in 2019 (1 year)
(4) "View synthesis deep network development for plenoptic images" funded by Korea Electronics and Telecommunications Research Institute, 2018~2020
(5) "CCTV security system for monitoring marine port" funded by Korea Institute of Ocean Science & Technology, 2019~2021 (3 years)
(6) "X-ray image quality improvement in container inspection systems" funded by Korea Institute of Ocean Science & Technology, 2019~2024 (5 years)
(7) "Material discrimination using dual energy X-ray images" funded by Korea Institute of Ocean Science & Technology, 2015~2019 (5 years)

- **Laboratory hardware capabilities**

Since our lab is an image processing and deep learning lab, we have recent PCs and workstations prepared specially for training of deep-learning models. We have a large number of PCs with multiple GPU units. GPU models range between NVIDIA's TITAN RTX, RTX2080 and RTX3090.

Problem Statement

In the real world, the visually impaired and blind people do not have the ability to go outside without a partner or assistant person. Sometimes those people have the desire to visit normal places like markets, shops, restaurants or other emergency places like hospital, pharmacy and police stations without depending on other persons. The visually impaired persons face a lot of problems in street navigation as it is difficult to navigate through the streets individually to reach their destination by walk or by public transportation depending on basic tools like blind sticks or common sense. The problem become even harder when it comes to street crossing, walking on platforms, obstacle and car avoidance or even reaching to the nearest bus station on time. Those people may get the assistance from a machine-vision AI-based navigation system to help them accurately navigate in the streets, save them from street car-accidents, and give them a compensation to the visual impairing by accurate voice orders. If visually impaired person follow voice command, he can safely and easily reach to his destination.

Proposed Solution

We propose an AI-based machine-vision hardware/software implementation to solve the problem of the individual street navigation for visually impaired individuals. With the aid of the current achievements of object detection and semantic segmentation, we can integrate those great computer vision algorithms in our solution to provide a micro-computer board a street-scene perception.

The recent semantic segmentation methods which are trained on the popular segmentation dataset "Cityscapes" provide a detailed street scene understanding via segmentation. This model can be easily integrated in an AI neural inference unit with camera like OAK-D or raspberry pi. Another tool should be integrated in our application which is the GPS which receive the desired location by the user, this GPS service can be implemented using mobile application which receive the voice command and send it to Google maps (or any alternative service), then the smart phone provide the directions to the user through a custom application, during the GPS navigation the OAK-D board is visualizing the street and provide the specific street crossing, platform positioning, obstacle avoidance command until user reach to his destination. The software also will include the ability to navigate to the nearest bus station with an optical character recognition (OCR) visual ability to insure the right bus number.

Also with the aid of object detection models, we can provide the ability of 3d object detection of basic street elements like vehicles, persons, bikes and etc... This 3d object detection can provide the information about distances using the camera depth data. This module can give accurate navigation commands like a description of the objects in the scene a distance level of the obstacles, cars and persons for better visual-navigation accuracy

Further Description

We provide briefly the hardware and software implementations of our proposed solution in the following subsection:

Hardware implementation:

The hardware implementation of our solution depends mainly on two boards:

1. OpenCV AI Kit (OAK-D).
2. Raspberry Pi (4 or zero).
3. Android Smart phone (we will try to implement an IOS version too but it is not confirmed).

The Opencv AI Kit (OAK-D) will be used as AI neural inference unit for deep learning semantic segmentation and object detection models. While the raspberry pi will be a side board which will be used as the host micro-computer board with Linux operating system, the raspberry pi 4 (or zero depending on the final solution dimensions) will be also used the to provide the Wi-Fi connection with smart phone and OAK-D board.

Smart phone mobile-application will be created to send the voice commands and receive the voice orders from user. The smart phone must have a GPS unit to provide the real-time position latitude and longitude which will be used to control the operation of OAK-D with the semantic segmentation and object detection inference models and which modes depending on the situation.

The whole system will be integrated to a compact-size tool to be used as a head mount tool on a hat or a cap. The head mount is the best way to fix the system as it will take the same tilt and pan angles of the face which make the system similar to the human eyes. Figure 1 shows a simple prototype for the head mount solution.

Figure 1. The proposed head mount solution

A block-diagram of the system is shown below in Figure 2.

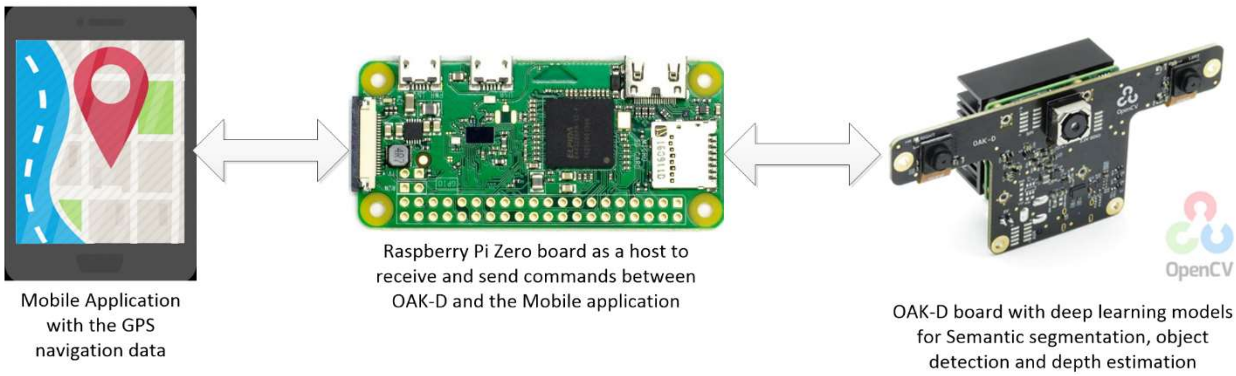

Figure 2. Block diagram of the system

<u>Software implementation:</u>

There will be three software parts in our project, as following:

<u>OAK-D deep learning models:</u>

In this software, we might use a light semantic segmentation model like MobileNetv2 based models which are built to run mobile device with limited resources. We will use such model to obtain a real-time processing on OAK-D. Also MobileNet-SSD can be used for object detection or can be combined with semantic segmentation to obtain instance segmentation if needed.

A depth estimation deep learning model can be used to obtain some 3D perception and accurate distance estimation. One can employ midasnet which is trained to estimate monocular depth using mixing datasets.

<u>Raspberry pi board:</u>

Raspberry pi board will be prepared with Linux-OS to be used as the host board of OAK-D using the usb connection. It will be responsible for taking the GPS position-based navigation commands through WI-FI to establish the deep learning models on the OAK-D board. Also, it will receive the visually-based navigation commands and send it to the mobile application through the WI-FI to be spoken to the user.

<u>Smart phone application:</u>

A mobile phone application will be designed and created especially for our solution. It will include Voice based navigation system. It will be responsible to send position data to raspberry pi board and receive the visually-based commands to be spoken to the user to provide the real-time street navigation.

Below is some scenarios of what the OAK-D board see and the corresponding command that user will hear in this situation.

Shown below, Figure 3, 4, 5 and 6 are some scenarios with the corresponding camera scene and the expected voice command in this case.

Figure 3. If the user is not on the platform, the OAK will detect that case and give him a voice command to go right until he reaches to the platform.

Figure 4. If the user has to cross the street, then the OAK will detect the traffic light for pedestrians and when it becomes green, it will give the voice command for crossing.

Figure 5. If the user is normally walking on the platform, then the OAK will detect trees, pedestrians, vehicles, and any other obstacles, then it will give the voice command for avoiding the this object.

Figure 6. If the user want to take the bus, then the OAK and GPS will give him the commands to the nearest station. By the using of text detection and recognition model it can verify the bus number and give the command to take the bus.

Limitation of the proposed solution

While the solution provide vision and position based navigation for visually impaired persons, the system has limitations because the camera has a fixed field of view (FOV) and it is head mounted so there is some dead-zone for the camera which it cannot see. So it is recommended to use this solution behind the tradition visually impaired stick such that the user is aware of the very close objects or suddenly appearing occluded objects in front of him which are not detected by the camera.

Image credits:

Figure 1: taken and edited from original: https://www.cksinfo.com/fashion/hats/index.html.

Figure 2: Three different images edited, originally: https://www.iconfinder.com/icons/3468771/android_map_gps_app_gps_navigation_map_application_navigation_app_icon.- https://vilros.com/products/raspberry-pi-zero-w- https://store.opencv.ai/products/oak-d

Figure 3, 4 and 5 are taken from cityscape dataset website and edited: https://www.cityscapes-dataset.com/examples/#fine-annotations

Figure 6: taken from Pascal VOC2009 dataset and edited: http://host.robots.ox.ac.uk/pascal/VOC/voc2009/index.html